

ReactHuman: A Physics-Grounded Benchmark for Human-Like Reactive Decision-Making in Embodied Multimodal LLMs

Yizhan Li^{1,2}, Jianxin You^{1,2}, Mengyang Xiong^{3,2}, Yinhan Chen³, Zicheng Zhao⁴, Dekun Wu^{1,2}, Dongqing Zhang⁵, Bang Liu^{1,2*}

¹Université de Montréal

²Mila – Quebec AI Institute

³McGill University

⁴McMaster University

⁵Meta Platforms

{yizhan.li, jianxin.you, dekun.wu, bang.liu}@umontreal.ca, {mengyang.xiong, yinhan.chen}@mail.mcgill.ca, zhaoz149@mcmaster.ca, zhangdq20@gmail.com

Abstract

Reacting to sudden physical hazards (catching a slipping plate, dodging a falling knife) is both a meaningful test of embodied intelligence and a hard requirement for deploying multimodal large language models (MLLMs) as the decision core of household robots. Existing evaluations, however, probe intuitive physics passively through question answering over videos, or target deliberate, long-horizon tasks such as navigation and rearrangement; none measure whether a model can turn physical understanding into immediate, safety-critical action. We introduce ReactHuman, the first physics-grounded benchmark for human-like reactive decision-making, in which the evaluated MLLM acts as the brain of a simulated humanoid facing sudden household hazards; it spans 17 event families and over 1,000 bit-for-bit reproducible scenes with exact, annotation-free ground truth derived from 240 Hz rigid-body simulation, including adversarial objects whose appearance contradicts their physics (a foam anvil, a steel apple). We further design a five-metric suite that scores each reaction along three axes: reasonable, safe, and physically grounded. We physically execute every committed plan so that decisions have observable consequences. With this harness we evaluate seven representative MLLMs. Results show that reactive safety is far from solved: models mishandle roughly one hazard in three, act from fixed dispositions rather than the observed scene, trust appearance over motion, and miss interception points at meter scale even when the chosen action is correct; none of these failures shrink with model scale. ReactHuman thus offers both a fine-grained diagnosis and a scalable training signal toward physically grounded, safety-aware embodied agents. The benchmark can be found here: <https://huggingface.co/datasets/Alan123/reacthuman-benchmark-scaled>

1 Introduction

Humans react to sudden physical hazards in a fraction of a second: we catch a slipping plate, dodge a falling knife, and brace against a toppling wardrobe. These reactions fuse *semantic* judgment (what is this object, and is it safe to touch?)

with *intuitive physics* (where and when will it arrive?). As multimodal large language models (MLLMs) are placed at the cognitive core of household robots and embodied assistants (Ahn et al. 2022; Driess et al. 2023), the same competence becomes a deployment requirement: we want an agent that reacts *like a competent person*, whose reaction is at once *safe* (it avoids harm), *reasonable* (it is the action a sensible human would choose and defend), and *physically grounded* (the hand actually meets the object in time). An agent that elects to catch a falling chef’s knife, or that decides correctly yet grasps half a meter wide, has failed in a way no captioning or VQA benchmark can reveal.

Existing evaluations do not measure this competence. Physical-reasoning benchmarks probe intuitive physics *passively*, through plausibility judgments or question answering over videos (Bear et al. 2021; Yi et al. 2020; Zheng et al. 2024; Chow et al. 2025), while embodied-AI benchmarks emphasize deliberate, long-horizon tasks such as navigation and rearrangement (Szot et al. 2021; Li et al. 2022). Neither reveals whether a model can convert physical understanding into an *immediate, safety-critical action* nor diagnose *why* it fails when it does: a model may misread the hazard, choose an unsafe action despite understanding it, or choose correctly yet mispredict the interception point by a meter.

We introduce **ReactHuman**, a benchmark that evaluates whether MLLMs react to sudden hazards *like a competent human* by making the MLLM the decision-making brain of a simulated humanoid that must actually carry the reaction out. We check if they act safely, reasonably, and with physically grounded movement in physically simulated indoor scenes. ReactHuman adopts a *freeze-and-predict* protocol: a full-physics simulation of a sudden household event runs to a critical decision moment and freezes; the model observes the unfolding event and must commit to a structured reaction plan: a walking command plus a hand-keyframe trajectory, from which the executed action (CATCH, DODGE, or No-ACTION) is derived. Because inference happens while simulated time is paused, models are compared on decision quality alone, uncontaminated by API latency. Because every scene is generated deterministically from a seed and labeled from simulator state, all scores are computed against

*Corresponding author.

exact physical ground truth. And because the committed plan is then *executed by a simulated humanoid* driven by a pre-trained RL walking policy (Rudin et al. 2022; Unitree Robotics 2024) in the same physics engine, every decision has an observable physical outcome (the catch connects or the toppling shelf strikes the agent) rather than remaining a multiple-choice answer, which is what separates ReactHuman from VQA-style physical-reasoning evaluation.

ReactHuman is built for *diagnosis*, not a single leaderboard number. Its **17 event families** (Figure 1) are chosen so that different families stress different links in the perception–reasoning–action chain: tabletop drops make the trajectory trivial and isolate semantic choice; pendulums and bouncing balls make the semantic choice easy and isolate interception; chain reactions require causal propagation before any kinematics; *adversarial objects* (a foam anvil, a steel apple, a lead-core tennis ball) decouple appearance from physics, so that the correct action is recoverable only from observed motion. Complementing the taxonomy, a **five-metric suite** instruments the three axes of a human-like reaction: semantic action accuracy and action–intent alignment (is the reaction *reasonable?*), safety validity (is it *safe?*), and physical end-point distance and hand-distance evolution (is it *physically grounded?*).

Our contributions are:

- **Benchmark.** A physics-grounded, freeze-and-predict evaluation of human-like reactive decisions along three axes: safety, reasonability, and physical grounding. The evaluation covers 17 sudden-event families, over 1000 reproducible scenes, three synchronized camera views, and exact simulator-derived ground truth (action, 3D impact point, time-to-floor), including adversarial appearance–physics probes, with every committed decision executed by a simulated humanoid in the same physics engine.
- **Generation pipeline.** A hybrid LLM-planned, seed-deterministic pipeline in which a frozen LLM contributes semantic diversity (objects, rooms, event routing from natural-language descriptions) while a procedural randomizer owns every physical parameter. This makes scenes bit-for-bit reproducible and the benchmark extensible past 10^4 scenes without human annotation.
- **Metric suite.** Five complementary metrics organized along the three axes: reasonability, safety, and physical grounding. This separates an unreasonable choice from an unsafe one from a kinematically ungrounded one, with an explicit safety-rule system and structured-output protocol.
- **Empirical study.** An evaluation of seven MLLMs across 306 scenes and all 17 families, revealing that safety failures concentrate where evasion is required, that action choice follows fixed per-model dispositions rather than the scene, that neither accuracy nor safety improves with model scale, and that models never revise appearance-based physics judgments from observed motion.

2 Related Work

Physical reasoning benchmarks. Physion and Physion++ (Bear et al. 2021; Tung et al. 2023) test forward predic-

tion of physical outcomes; CLEVRER (Yi et al. 2020) and IntPhys (Riochet et al. 2018) probe causal and violation-of-expectation reasoning; ContPhy (Zheng et al. 2024) extends to deformables and fluids. For modern MLLMs, PhysBench (Chow et al. 2025) is the closest neighbor: it measures physical-world understanding via multiple-choice VQA over a model population that largely overlaps ours. Physics-IQ (Motamed et al. 2025) scores video-generation models for physical plausibility. ReactHuman differs on two axes simultaneously: the model’s output is an *action with safety consequences* rather than an answer, and ground truth is generated automatically and exactly by simulation, so the quantitative track scales.

Embodied-AI and humanoid benchmarks. Habitat (Savva et al. 2019; Szot et al. 2021), AI2-THOR (Kolve et al. 2017), ThreeDWorld (Gan et al. 2021), ManiSkill (Mu et al. 2021), ManiSkill2 (Gu et al. 2023), BEHAVIOR-1K (Li et al. 2022), and OpenEQA (Majumdar et al. 2024) evaluate navigation, manipulation, rearrangement, or situated QA on deliberate time scales; HumanoidBench (Sferrazza et al. 2024) targets whole-body *control*. None evaluate immediate reactive safety. ReactHuman isolates the *decision layer* while retaining embodiment: following a brain–spine–body decoupling, the evaluated MLLM acts as the brain of a simulated humanoid whose balance and joint torques are delegated to a pre-trained whole-body controller (Rudin et al. 2022; Unitree Robotics 2024), so scores are attributable to reasoning rather than policy quality, yet every decision is still physically executed in scene.

MLLMs as embodied decision modules vs. VLAs. One line of work deploys frozen MLLMs as zero-shot planners over discrete skills (Ahn et al. 2022; Driess et al. 2023); another fine-tunes vision–language backbones end-to-end into vision–language–action (VLA) policies emitting low-level control (Brohan et al. 2023; Kim et al. 2024; Black et al. 2024). ReactHuman’s harness spans the two: coupling a frozen MLLM to a pre-trained whole-body controller and a simulated humanoid turns any off-the-shelf MLLM into a zero-shot vision-to-action agent, without the action fine-tuning that defines VLAs. We evaluate general-purpose MLLMs because they are what is deployed today as the reasoning core of embodied systems; native VLA policies can be dropped into the same scenes, ground truth, and metrics unchanged—a comparison we leave to future work.

Procedural and LLM-assisted scene generation. Kubric (Greff et al. 2022) established scalable simulator–renderer tooling; Objaverse (Deitke et al. 2023) supplies large-scale assets; recent systems use LLMs to compose environments and tasks (Wang et al. 2024; Yang et al. 2024). ReactHuman combines the two: an LLM performs *semantic* planning from natural language while a deterministic, seeded randomizer retains exclusive control of *physical* parameters, preserving the reproducibility and label exactness that fully LLM-driven generation lacks.

3 The ReactHuman Benchmark

3.1 Freeze-and-Predict Protocol

Each episode places a virtual observer. A digital human standing in an indoor room is required to react to a sudden physical event at t_0 : an object begins to fall, slide, roll, topple, swing, or be struck toward the observer. The model receives the *observation window*: frames from t_0 to $t_0 + \Delta$ (default $\Delta \approx 0.6$ s, before the outcome is visually resolved) from up to three synchronized viewpoints (standing observer, close-up at the event origin, overhead). Simulated time then freezes and the model must output a structured reaction plan

$$\text{output} = \{\text{intent}, \text{confidence}, \text{walking_cmd}, \text{keyframes}\}, \quad (1)$$

where *intent* is free-form reasoning, *confidence* $\in [0, 1]$, *walking_cmd* is a base-velocity command, and *keyframes* $\{\mathbf{h}_1, \dots, \mathbf{h}_n\}$, $\mathbf{h}_i \in \mathbb{R}^3$, is the predicted hand trajectory. Freezing time makes the decision problem identical for every model and removes inference latency as a confound; scoring is fully deterministic.

Humanoid execution. The reaction plan is not merely graded on paper: simulation then resumes and the plan is *executed by a simulated humanoid* (a Unitree G1) placed in the same Genesis scene, following a brain–spine–body decoupling: the evaluated MLLM is the *brain*; a pre-trained whole-body controller (the *spine*) converts the walking command and hand keyframes into balance and joint torques; the humanoid *body* interacts with the falling object under full rigid-body physics. Each episode therefore yields an execution video in which the chosen reaction has observable physical consequences. The catch connects or misses, the dodge clears the impact zone or fails to. This distinguishes ReactHuman from passive VQA: the unit of evaluation is an embodied interaction, not an answer. For metric *scoring* we use the simulator-derived ground truth (§4), which keeps scores deterministic and independent of controller quality; execution provides outcome-level verification. Representative rollouts are shown in Figure 3.

Action space. The model does not pick from a menu. It outputs a motor plan, a base-velocity walking command plus hand keyframes, and a deterministic rule classifies the plan into one of three primitives: EXECUTE_CATCH (walk toward the object and reach for it), TRIGGER_DODGE (move clear of its path), or NO_ACTION (stay put, hands at rest). We score what the body would actually do, not what the model claims, so a model cannot pass by simply saying the right word. Ground-truth labels follow object properties and event kinematics, not appearance: light, graspable, benign objects are labeled CATCH; sharp, hot, shattering, heavy, or fast objects are labeled DODGE. NO_ACTION is correct only in the few scenes where the event cannot reach the observer, and choosing it under an active hazard counts as a safety violation. On the evaluated set the labels split 141/156/9 across CATCH/DODGE/NO_ACTION, so majority-class guessing (always DODGE) attains 51.0%.

3.2 A Taxonomy of 17 Sudden-Event Families

Figure 1 summarizes the families. Beyond covering distinct dynamics (free fall, friction-driven sliding, inverted-pendulum toppling, hinge-constrained rotation, ballistic flight with restitution, momentum transfer), families differ in *decision structure*. In *object_drop* the trajectory is trivial and the decision is dominated by object semantics. In *pendulum_swing* and *bouncing_object* semantics are easy but the interception point is time-varying or multimodal. In *chain_reaction* and *multi_object* the threatening object is initially *stationary*: the model must propagate causality (A will strike B; momentum will travel down the row) before any kinematic estimate is possible. This factorization lets ReactHuman localize failures to perception, causal reasoning, semantic judgment, or spatial calculation.

3.3 Adversarial Appearance–Physics Probes

The 82-object asset library contains 14 *adversarial variants* whose visual identity contradicts their physical parameters: a foam anvil (looks cast-iron; light and safe to catch), a steel apple (looks like fruit; arrives with bruising momentum), a lead-core tennis ball, a foam mirror, a hollow prop door, a foam ceiling panel, a lead sandbag, a solid-steel can. For these objects the appearance prior implies the *wrong* action; the correct one is recoverable only from observed dynamics in the early frames. Section 5 probes these objects directly: across 280 decisions no model ever questions an object’s material or weight, and disguised objects receive exactly the treatment of their genuine look-alikes.

3.4 Scene Generation and Ground Truth

Every scene is produced by a three-stage pipeline (Figure 2).

Stage 1: LLM semantic planning. A frozen LLM parses a natural-language description (“a cast-iron pan slides off the counter while someone is cooking”) into a validated selection: event family, object (from a per-family catalogue), room type, and family-specific hints (e.g., which intermediate surface a cascading object lands on). Outputs are validated against the catalogue with automatic retry on malformed responses. Objects named in descriptions but absent from the library can be sourced automatically from Objaverse (Deitke et al. 2023): candidates retrieved by LVIS label are scored for shape plausibility, re-oriented, welded, and registered.

Stage 2: Seeded physical randomization. A deterministic randomizer, a pure function of an integer seed, samples every physical and visual parameter: placement and overhang, initial linear and angular velocities, friction and restitution, room and table geometry offsets, lighting, and camera poses (with a retry loop guaranteeing the object remains in frame). Family-specific samplers encode each event’s physics: ramp angles are floored above the friction cone so sliding is guaranteed; ladder lean angles bifurcate into slide-down versus tip-over regimes; chain-reaction targets are settled to rest and only then struck by the trigger. Physics is never scripted mid-scene; outcomes emerge from initial conditions alone.

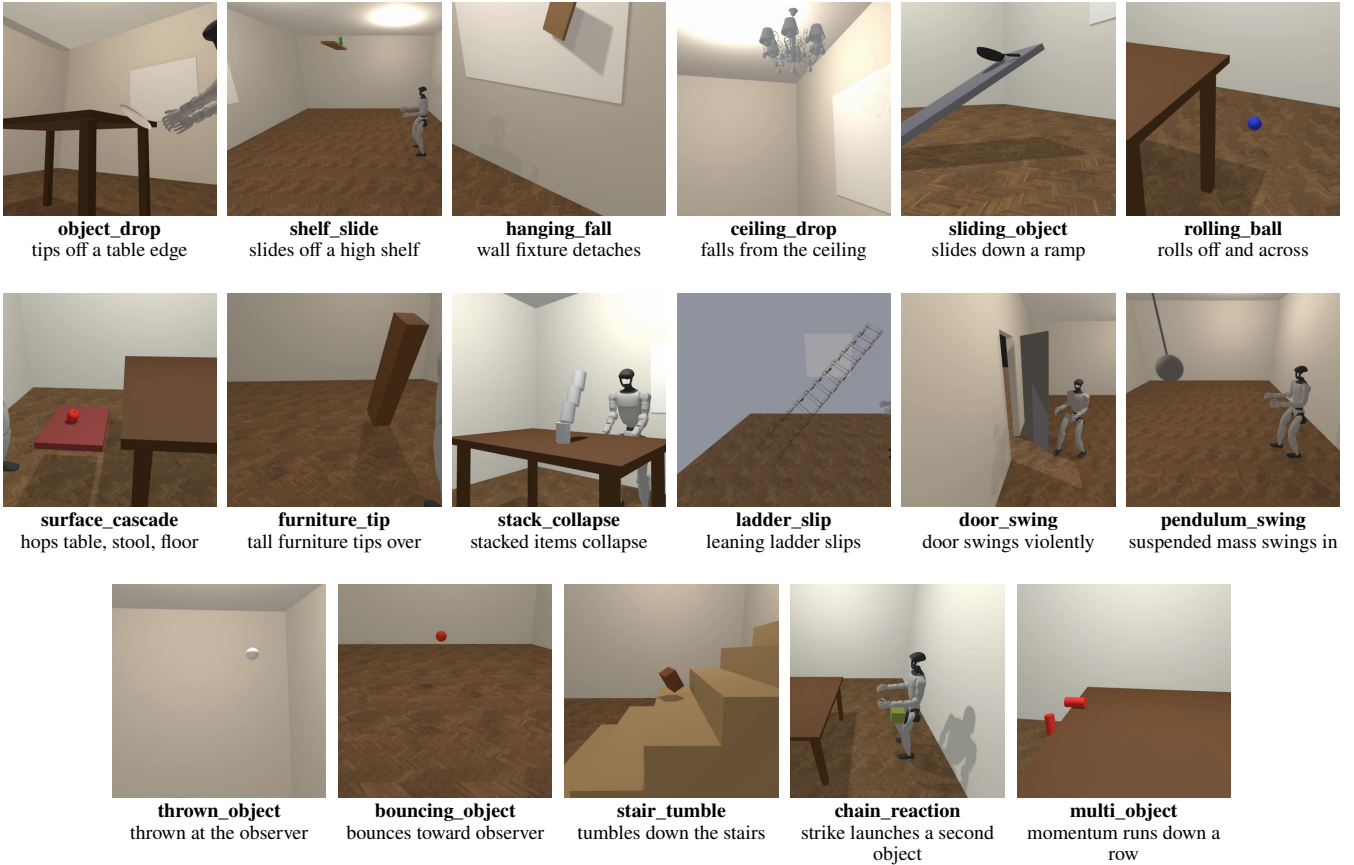


Figure 1: **The 17 sudden-event families of ReactHuman.** The families span six motion archetypes (fall, slide & roll, topple, articulated, ballistic, and causal chain); each is evaluated on 18 scenes in §5, and every thumbnail is a real benchmark scene.

Stage 3: Simulation and labeling. Scenes are simulated in Genesis (Genesis Authors 2024) with a rigid-body solver at 240 Hz and rendered at $1280 \times 720 / 60$ fps from three cameras. Ground truth is extracted from simulator state: the action label from the object’s catalogued properties and event kinematics; the impact point $\mathbf{p}_{\text{impact}} \in \mathbb{R}^3$ where the trajectory crosses the observer’s reach plane; and time-to-floor from first floor contact. Each scene ships its complete specification; re-running the specification reproduces the simulation bit-for-bit, and geometric validators check spec-simulation consistency before rendering.

This division of labor is deliberate: the LLM contributes semantic diversity and natural-language grounding but never touches numbers; the seeded randomizer owns all continuous parameters, so labels are exact and the benchmark extends procedurally past 10^4 scenes (with adversarial variants injected at a configurable rate) at zero annotation cost.

4 Metric Suite

We report five metrics per decision, each aimed at a different kind of failure: a model can pick the right action but still be unsafe, or give a sensible reason yet send its hands to the wrong place. Throughout, s indexes scenes and m models. The model never outputs an action label; we recover

the action from its motor plan (the walking command and hand keyframes) with a fixed rule that is released with the evaluation code.

1. Semantic Action Accuracy (SAA). $\text{SAA}(s, m) = 1[\hat{a}(s, m) = a^*(s)]$: whether the chosen action matches the ground truth. This checks the action label alone, so a model can score $\text{SAA} = 1$ and still break a safety rule.

2. Safety Validity. A binary flag for safety-critical mistakes, scored independently of SAA. A decision is unsafe if it breaks any of four rules: (R1) no parseable output; (R2) CATCH on an object labeled dangerous; (R3) No_ACTION while danger is present; (R4) CATCH or No_ACTION when the ground truth is DODGE. We record which rule was broken. We keep safety separate from SAA on purpose: catching a dangerous object is unsafe even when catching is a reasonable move in general.

3. Physical Endpoint Distance. $d_{\text{end}}(s, m) = \|\mathbf{h}_{\text{final}}(s, m) - \mathbf{p}_{\text{impact}}(s)\|_2$ in meters, the distance between the final hand position and the true impact point. A model can choose the right action and still put its hands in the wrong place.

4. Action-Intent Alignment (AIA). A three-level score (0, 0.5, 1) for whether the stated `intent` matches the goal of

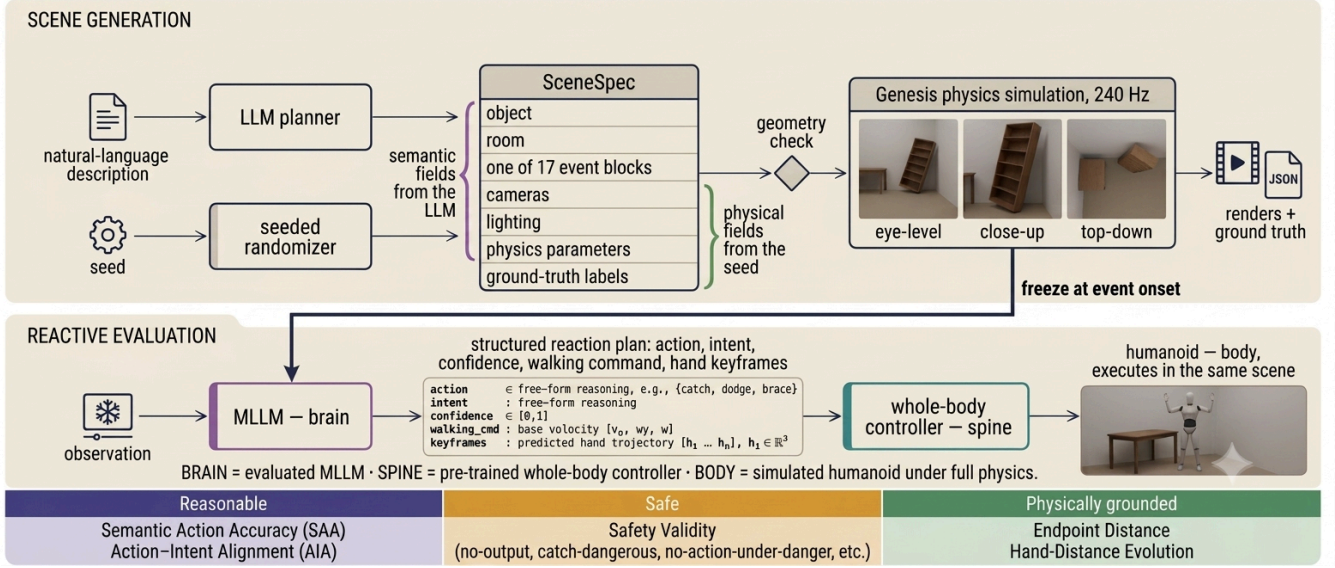


Figure 2: **The ReactHuman pipeline.** A frozen LLM planner and a seeded randomizer jointly populate a single SceneSpec. The LLM sets semantics (object, room, event family), and the seed sets all physics while Genesis simulates at 240 Hz into three synchronized views with simulator-derived ground truth. At a frozen observation, the evaluated MLLM (the brain) commits a structured reaction plan that a pre-trained controller (spine) executes on a simulated humanoid (body) in the same scene. The plan is scored on three axes: reasonable, safe, and physically grounded.

the ground-truth action. It separates a real misunderstanding from a right idea with a bad output: high AIA with $SAA = 0$ means the model understood the scene but wrote the wrong plan, while low AIA means it misread the scene. Intent is currently scored by keyword matching.

5. Hand-Distance Evolution (HDE). Along the keyframe path we compute $d_i = \|\mathbf{h}_i - \mathbf{p}_{\text{impact}}\|_2$, and report the closest the hand gets, $d_{\min} = \min_i d_i$, together with how much it closes in, $d_1 - d_{\min}$. This shows whether the hand actually moves toward the target, which the endpoint alone can hide: a path can wander and still finish near the impact point.

Aggregation. We average each metric per model and per family, dropping missing values from the denominator and reporting how many. Safety takes priority in our plots: once a decision breaks a safety rule, that dominates its rating regardless of the other numbers.

5 Experiments

5.1 Setup

Models. We evaluate seven MLLMs zero-shot through one unified API (OpenRouter): five frontier models (Claude Opus 4.8, GPT-5.5, Gemini 2.5 Flash, Kimi K2.6, Qwen3-VL-235B) and two small open-weight models (Qwen3-VL-30B and Gemma-3-27B), which probe whether the benchmark separates capability tiers.¹ All models receive identical

¹Model IDs: anthropic/claude-opus-4.8, openai/gpt-5.5, google/gemini-2.5-flash, moonshotai/kimi-k2.6, qwen/qwen3-vl-235b-a22b-instruct, qwen/qwen3-vl-30b-a3b-

prompts and output schema, observe the same ~ 0.6 s multi-view window, and answer as JSON.

Controller. The legs follow a pre-trained RL walking policy (Rudin et al. 2022; Unitree Robotics 2024); a scripted upper-body layer tracks the commanded hand keyframes. Early prototypes used motion-imitation whole-body controllers (Luo et al. 2023; Tessler et al. 2024), which proved harder to steer with task-level commands and less stable under sudden impacts, so we kept the simpler decoupled spine.

Scenes. We evaluate a balanced subset of 306 scenes, 18 per family across all 17 families, so that no family dominates the aggregate metrics. Paired with the seven models this yields 2,138 decisions (four lost to API errors).

5.2 Results and Analysis

Key findings. Table 1 reports the per-model results; three patterns stand out. **(1) Models fail most when they need to evade.** On scenes whose correct response is DODGE, 35.9% of decisions violate a safety rule, and this holds for every model (23–66%). The most common error is simply not moving: 253 of the 392 mistakes are freezing in place. **(2) Each model has a fixed bias.** Six of seven models prefer to retreat and score much better on dodge scenes than on catch scenes (Figure 4); Claude is the opposite, catching well (67%) but engaging hazards it should avoid (34% on dodge scenes). No model adapts its behavior to the scene. **(3) Bigger is not safer.** The 27B Gemma ranks second on accuracy (62.4%, just behind GPT-5.5) and first on safety (88.2%), and both ~ 30 B open-weight models outscore most of the frontier tier

instruct, google/gemma-3-27b-it.

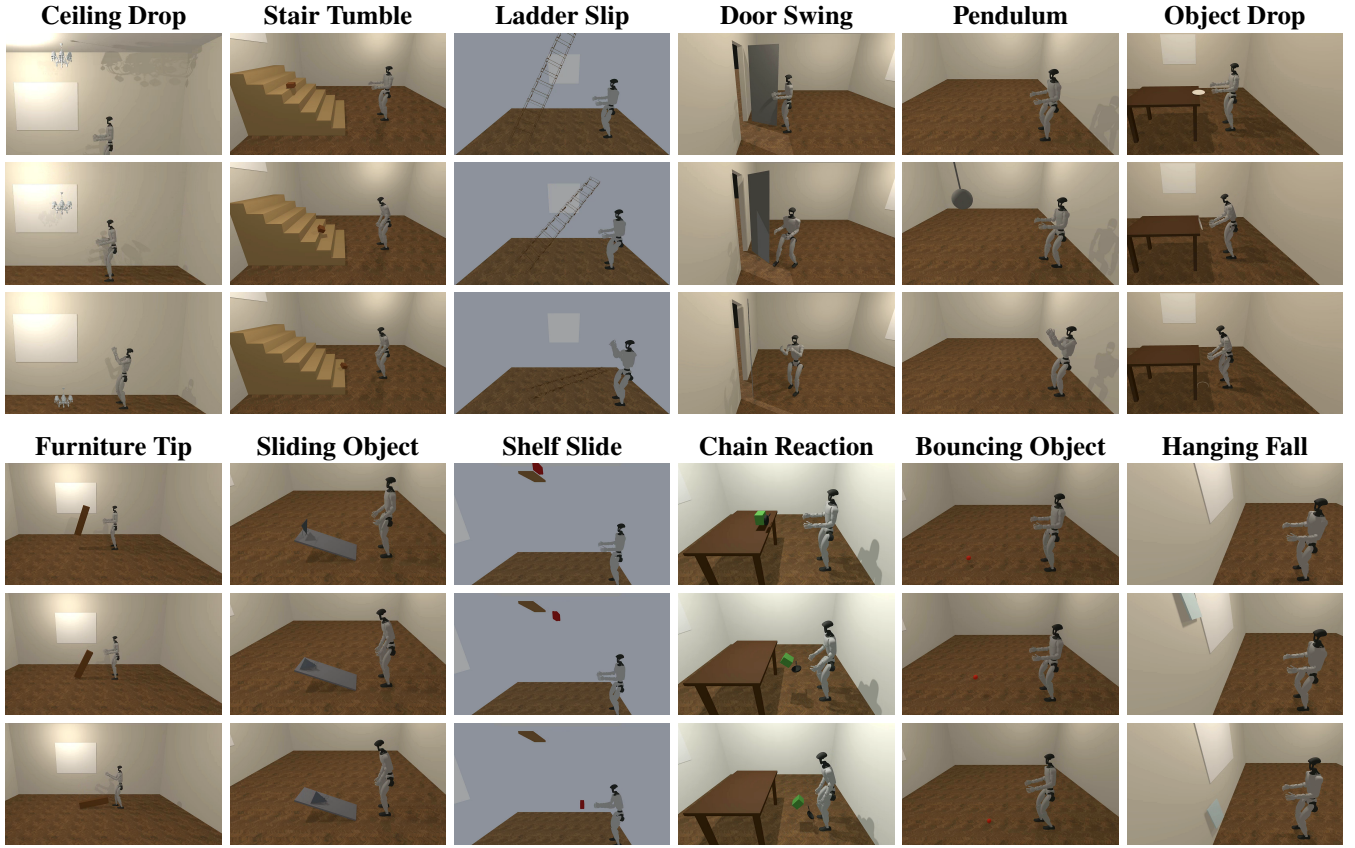


Figure 3: **Twelve scenes from twelve event families.** Each column is one scene read top to bottom (three frames, third-person view), with the event family named above; all reactions shown are produced by Claude Opus 4.8. Every decision is physically executed, so each reaction ends in an observable outcome.

Model	SAA $\uparrow$	Safety $\uparrow$	$d_{\text{end}}(\text{m}) \downarrow$	AIA $\uparrow$	$d_{\text{min}}(\text{m}) \downarrow$
Claude Opus 4.8	50.7	64.7	1.07	0.41	1.06
GPT-5.5	63.7	86.3	1.23	0.47	1.21
Gemini 2.5 Flash	48.7	78.1	1.28	0.47	1.28
Kimi K2.6	40.9	78.2	1.30	0.32	1.28
Qwen3-VL-235B	56.4	87.2	1.25	0.48	1.23
Qwen3-VL-30B	54.9	83.0	1.20	0.50	1.17
Gemma-3-27B	62.4	88.2	1.26	0.61	1.25
Mean	54.0	80.8	1.23	0.47	1.21

Table 1: Main results (%; distances in meters) over 306 scenes spanning all 17 families (2,138 scene–model decisions; four lost to API errors). Distance metrics are computed on scenes with a defined impact point; best per column in **bold**.

on both columns, while the strongest flagship, Claude Opus 4.8, is the least safe model in the table (64.7%).

Per-family breakdown. Since six of seven models favor retreat (Finding 2), families whose answer is DODGE score high for free; we therefore compare catch and dodge families separately. Catch families get harder as more pieces move: a single bouncing ball is caught 66.7% of the time, a col-

lapsing stack of cans only 38.9%. Dodge families get harder as warning time shrinks: a slowly falling ceiling fixture is evaded 91.3% of the time, an object thrown at the robot only 54.8%, and nearly half of the thrown-object decisions (44%) break a safety rule.

Safety errors. Of the 410 violations, freezing under danger is the largest class (R3, 162), ahead of staying put or engaging when evasion was required (R4, 128) and catching an object labeled dangerous (R2, 120); R1 (unparseable output) never fired. Models fail by misjudging the hazard, not by lacking the right action.

Correct but out of reach. Saying CATCH is not catching. Even when models pick the right action, their hands land a median 0.48 m from where the object actually arrives, and 89% of the misses fall short. The cause is simple: models reach with the arms but do not step forward. In half of these decisions (49%) the robot stands more than 0.8 m away, farther than its arm can reach at all. A multiple-choice benchmark would mark every one of these correct, while physical execution shows they miss. This is why ReactHuman executes every plan.

Speed is perceived as binary. Twelve of the seventeen families sweep the object’s true speed across four bands

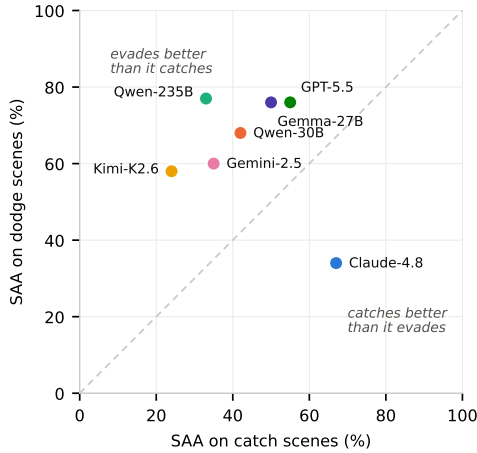


Figure 4: Action choice reflects the model, not the scene. Each point is one model: SAA on catch scenes (x) vs. dodge scenes (y); the dashed diagonal marks equal skill. Six models sit far above it, Claude far below, and the upper right, strong on both sides, stays empty.

while holding everything else fixed; in the remaining five gravity sets the pace, so we exclude them here. On these 826 decisions, models read speed only as moving-or-not (Figure 5a): at the slowest band they freeze twice as often (42% vs. roughly 21%), treating slow hazards as no event at all even when the motion is plainly visible in their input (Figure 5b), and accuracy drops with them (38.1% SAA vs. 47–54% at the faster bands). From normal to very fast the action distribution barely moves. Individual decisions are also unstable: a model keeps the same answer across all four speeds in only 27% of scene groups, and the slowest and fastest variants agree just 54% of the time. Models detect *that* something moves; *how fast* never enters the decision.

Voting does not help. The seven models agree unanimously on only 16% of scenes, and when they do, they are right 80% of the time. Pooling them buys little: a majority vote over all seven scores 62.1%, better than the average model (54.0%) but no better than the best single model (63.7%). They share blind spots, and an ensemble cannot vote shared errors away.

Adversarial probes. Forty scenes plant an object whose appearance lies about its physics: foam doors and foam ceiling panels disguised as heavy, lead sandbags and steel cans disguised as light. The disguise works completely: across the $40 \times 7 = 280$ decisions, no model ever voices a doubt about an object’s material or weight. For example, the foam ceiling panels are dodged in 49 of 56 decisions, just as real slabs would be, and the solid-steel cans get no discount for their weight: models reach for them exactly as often as for genuinely light cans (39% vs. 39% of decisions). For current MLLMs, seeing is believing: physics is assigned by visual category and never revised from observed motion.

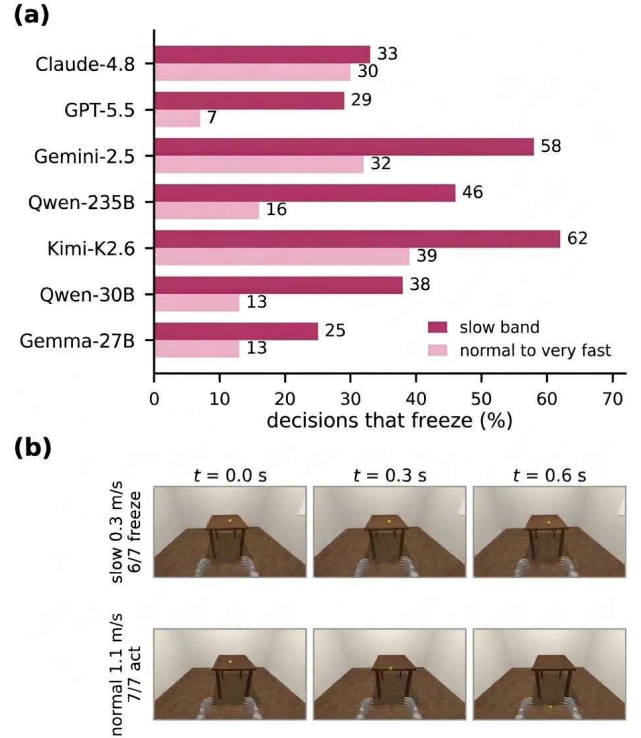


Figure 5: Slow hazards are missed by every model. (a) Freeze rate at the slowest speed band vs. the three faster bands, over the twelve families that genuinely sweep object speed: every model freezes more on slow hazards (pooled, 42% vs. 21%). (b) Claude’s input for one rolling-ball scene at two bands (ground truth CATCH in both). The slow ball still rolls toward the table edge and arrives 1 s after the window closes, yet six of seven models freeze, Claude included; at the normal band none freeze and Claude catches.

6 Discussion and Limitations

What ReactHuman reveals. The failures are not exotic: models freeze in the path of hazards they plainly see, apply one fixed disposition to every scene instead of reading it, treat speed as a binary and slow hazards as no event, and never question an object’s physics when its appearance disagrees. None of these improve with scale, and none are visible to an answer-only protocol. For deployment each points to a different mitigation: safety-constrained decoding for the freeze default, paired catch-and-dodge training to break fixed dispositions, and motion-grounded supervision for speed and material.

Limitations. Ground truth is simulator-derived; rigid-body simulation omits deformation and shattering, and conclusions should be validated on real hardware. The *decision* is open-loop by construction (one frozen observation, one

committed plan); the humanoid execution that follows is physically simulated but not re-planned mid-motion. Fully closed-loop evaluation with repeated observation–action cycles, cloth and fluid events, and native VLA policies are natural extensions. Finally, the evaluated models are moving targets: results reflect the specific API snapshots queried, and the standing benchmark, not any one number, is the contribution.

7 Conclusion

ReactHuman reframes physical-reasoning evaluation from passive question answering to reactive, safety-critical decision-making. Its 17 event families, exact simulator ground truth, adversarial appearance–physics probes, and five-metric diagnostic suite jointly measure not just whether an embodied MLLM *knows* physics, but whether it can act safely within it, and localize the failure when it cannot. All scenes are bit-for-bit reproducible and the LLM-planned, seed-deterministic pipeline extends the benchmark without annotation cost. Across seven models, reactive safety proves far from solved: models mishandle roughly one hazard in three, act from a fixed disposition rather than the scene in front of them, and do not improve with scale. We release the dataset to support further study.

Acknowledgments

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References

- Ahn, M.; Brohan, A.; Brown, N.; Chebotar, Y.; Cortes, O.; David, B.; et al. 2022. Do As I Can, Not As I Say: Grounding Language in Robotic Affordances. In *CoRL*.
- Bear, D. M.; Wang, E.; Mrowca, D.; Binder, F. J.; Tung, H.-Y. F.; et al. 2021. Physion: Evaluating Physical Prediction from Vision in Humans and Machines. In *NeurIPS Datasets and Benchmarks*.
- Black, K.; Brown, N.; Driess, D.; Esmail, A.; Equi, M.; Finn, C.; et al. 2024. π_0 : A Vision-Language-Action Flow Model for General Robot Control. *arXiv preprint arXiv:2410.24164*.
- Brohan, A.; Brown, N.; Carbajal, J.; Chebotar, Y.; Chen, X.; Choromanski, K.; et al. 2023. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. In *CoRL*.
- Chow, W.; Mao, J.; Li, B.; Seita, D.; Guizilini, V.; and Wang, Y. 2025. PhysBench: Benchmarking and Enhancing Vision-Language Models for Physical World Understanding. In *ICLR*.
- Deitke, M.; Schwenk, D.; Salvador, J.; Weihs, L.; Michel, O.; VanderBilt, E.; Schmidt, L.; Ehsani, K.; Kembhavi, A.; and Farhadi, A. 2023. Objaverse: A Universe of Annotated 3D Objects. In *CVPR*.
- Driess, D.; Xia, F.; Sajjadi, M. S. M.; Lynch, C.; Chowdhery, A.; Ichter, B.; et al. 2023. PaLM-E: An Embodied Multimodal Language Model. In *ICML*.
- Gan, C.; Schwartz, J.; Alter, S.; Mrowca, D.; Schrimpf, M.; Traer, J.; et al. 2021. ThreeDWorld: A Platform for Interactive Multi-Modal Physical Simulation. In *NeurIPS Datasets and Benchmarks*.
- Genesis Authors. 2024. Genesis: A Generative and Universal Physics Engine for Robotics and Beyond. <https://github.com/Genesis-Embodied-AI/Genesis>.
- Greff, K.; Belletti, F.; Beyer, L.; Doersch, C.; Du, Y.; Duckworth, D.; et al. 2022. Kubric: A Scalable Dataset Generator. In *CVPR*.
- Gu, J.; Xiang, F.; Li, X.; Ling, Z.; Liu, X.; Mu, T.; et al. 2023. ManiSkill2: A Unified Benchmark for Generalizable Manipulation Skills. In *ICLR*.
- Kim, M. J.; Pertsch, K.; Karamcheti, S.; Xiao, T.; Balakrishna, A.; Nair, S.; et al. 2024. OpenVLA: An Open-Source Vision-Language-Action Model. *arXiv preprint arXiv:2406.09246*.
- Kolve, E.; Mottaghi, R.; Han, W.; VanderBilt, E.; Weihs, L.; Herrasti, A.; Gordon, D.; Zhu, Y.; Gupta, A.; and Farhadi, A. 2017. AI2-THOR: An Interactive 3D Environment for Visual AI. *arXiv preprint arXiv:1712.05474*.
- Li, C.; Zhang, R.; Wong, J.; Gokmen, C.; Srivastava, S.; Martín-Martín, R.; et al. 2022. BEHAVIOR-1K: A Benchmark for Embodied AI with 1,000 Everyday Activities and Realistic Simulation. In *CoRL*.
- Luo, Z.; Cao, J.; Kitani, K.; Xu, W.; et al. 2023. Perpetual Humanoid Control for Real-time Simulated Avatars. In *ICCV*.
- Majumdar, A.; Ajay, A.; Zhang, X.; Putta, P.; Yenamandra, S.; et al. 2024. OpenEQA: Embodied Question Answering in the Era of Foundation Models. In *CVPR*.
- Motamed, S.; Culp, L.; Swersky, K.; Jaini, P.; and Geirhos, R. 2025. Do Generative Video Models Understand Physical principles? *arXiv preprint arXiv:2501.09038*.
- Mu, T.; Ling, Z.; Xiang, F.; Yang, D.; Li, X.; Tao, S.; Huang, Z.; Jia, Z.; and Su, H. 2021. ManiSkill: Generalizable Manipulation Skill Benchmark with Large-Scale Demonstrations. *arXiv:2107.14483*.
- Riochet, R.; Castro, M. Y.; Bernard, M.; Lerer, A.; Fergus, R.; Izard, V.; and Dupoux, E. 2018. IntPhys: A Framework and Benchmark for Visual Intuitive Physics Reasoning. *arXiv preprint arXiv:1803.07616*.
- Rudin, N.; Hoeller, D.; Reist, P.; and Hutter, M. 2022. Learning to Walk in Minutes Using Massively Parallel Deep Reinforcement Learning. In *Conference on Robot Learning*, 91–100.
- Savva, M.; Kadian, A.; Maksymets, O.; Zhao, Y.; Wijmans, E.; Jain, B.; Straub, J.; Liu, J.; Koltun, V.; Malik, J.; Parikh, D.; and Batra, D. 2019. Habitat: A Platform for Embodied AI Research. In *ICCV*.
- Sferrazza, C.; Huang, D.-M.; Lin, X.; Lee, Y.; and Abbeel, P. 2024. HumanoidBench: Simulated Humanoid Benchmark for Whole-Body Locomotion and Manipulation. In *RSS*.
- Szot, A.; Clegg, A.; Undersander, E.; Wijmans, E.; Zhao, Y.; Turner, J.; et al. 2021. Habitat 2.0: Training Home Assistants to Rearrange their Habitat. In *NeurIPS*.

Tessler, C.; Guo, Y.; Nabati, O.; Chechik, G.; and Peng, X. B. 2024. MaskedMimic: Unified Physics-Based Character Control Through Masked Motion Inpainting. *ACM Transactions on Graphics*, 43(6).

Tung, H.-Y. F.; Ding, M.; Chen, Z.; Bear, D.; Gan, C.; Tenenbaum, J. B.; Yamins, D. L.; Fan, J.; and Smith, K. 2023. Physion++: Evaluating Physical Scene Understanding that Requires Online Inference of Different Physical Properties. In *NeurIPS*.

Unitree Robotics. 2024. Unitree RL GYM: Reinforcement Learning Environments for Unitree Robots. https://github.com/unitreerobotics/unitree_rl_gym.

Wang, Y.; Xian, Z.; Chen, F.; Wang, T.-H.; Wang, Y.; Fragkiadaki, K.; Erickson, Z.; Held, D.; and Gan, C. 2024. RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation. In *ICML*.

Yang, Y.; Sun, F.-Y.; Weihs, L.; VanderBilt, E.; Herrasti, A.; Han, W.; et al. 2024. Holodeck: Language Guided Generation of 3D Embodied AI Environments. In *CVPR*.

Yi, K.; Gan, C.; Li, Y.; Kohli, P.; Wu, J.; Torralba, A.; and Tenenbaum, J. B. 2020. CLEVRER: Collision Events for Video Representation and Reasoning. In *ICLR*.

Zheng, Z.; Yan, X.; Chen, Z.; Wang, J.; Lim, Q. Z. E.; Tenenbaum, J. B.; and Gan, C. 2024. ContPhy: Continuum Physical Concept Learning and Reasoning from Videos. In *ICML*.